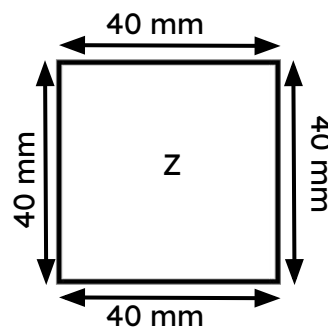
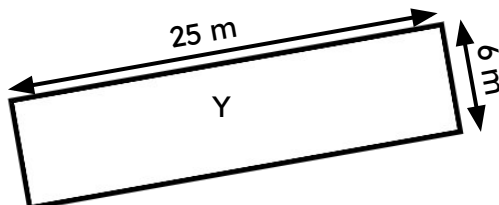
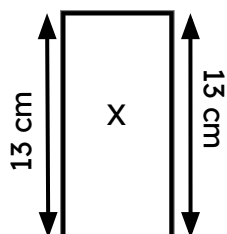




## Securing understanding of perimeter and area

### Task A: Calculating the perimeter and area of rectangles

1) Jun wants to work out the **area** and **perimeter** of each of these rectangles.



a) Which rectangles have “enough”, “too much”, or “not enough” information?

*X: Not enough, need the length of a shorter side*

*Y: The right amount, the other sides can be deduced*

*Z: Too much, need two non-parallel sides*

b) Find the **area** and **perimeter** of the rectangles where possible.

*For shape Y*

$$\begin{aligned} \text{Perimeter} &= 25 + 25 + 6 + 6 \\ &= 62 \text{ m} \end{aligned} \qquad \begin{aligned} \text{Area} &= 25 \times 6 \\ &= 150 \text{ m}^2 \end{aligned}$$

*For shape Z*

$$\begin{aligned} \text{Perimeter} &= 40 + 40 + 40 + 40 \\ &= 160 \text{ mm} \end{aligned} \qquad \begin{aligned} \text{Area} &= 40 \times 40 \\ &= 1600 \text{ mm}^2 \end{aligned}$$

c) Jun works out the **area** of Z using the calculation  $40 \times 40 \times 40 \times 40$ .  
What have they done wrong?

*They have multiplied every length together instead of multiplying the length by the width.*



## Task B: Identifying a rectangle from its perimeter

1) Laura wants to buy a plot of land to build a house. The table below shows some information about the plots.

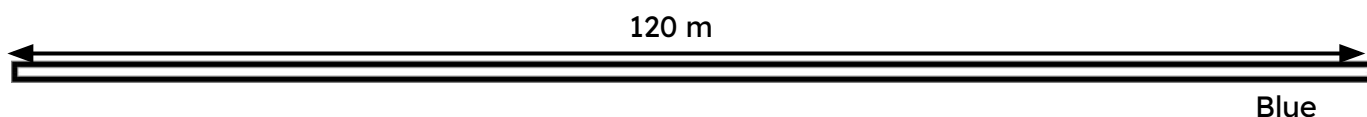
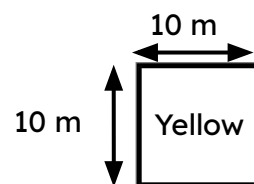
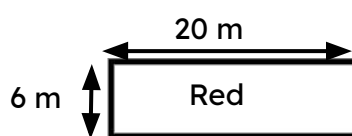
| Plot | Area (m <sup>2</sup> ) | Perimeter (m) |
|------|------------------------|---------------|
| X    | 100                    | 40            |
| Y    | 120                    | 242           |
| Z    | 120                    | 52            |

a) Which plot do you think they should choose?

*Any plot may be selected here*

By the table are plan views of the three plots.

| Plot | Area (m <sup>2</sup> ) | Perimeter (m) |
|------|------------------------|---------------|
| X    | 100                    | 40            |
| Y    | 120                    | 242           |
| Z    | 120                    | 52            |



b) Match each image to a row in the table.

*X = Yellow; Y = Blue; Z = Red*

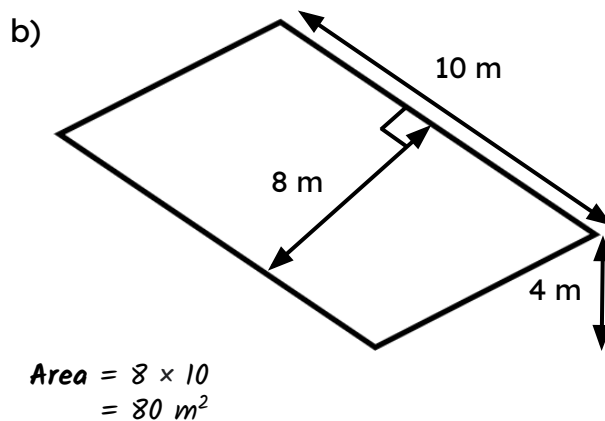
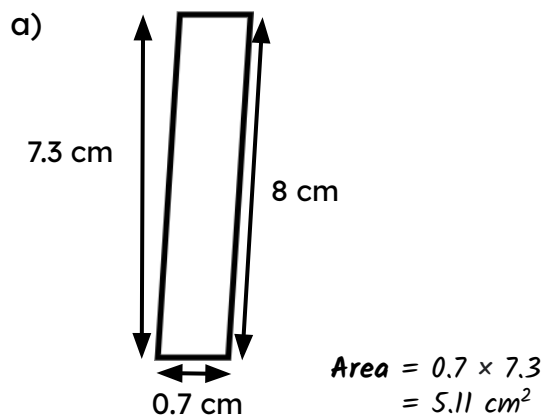
c) Explain whether you wish to change your answer to part A.

*Answers will vary here depending on the original plot chosen*



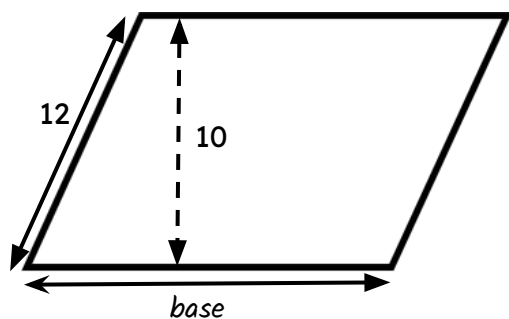
## Task C: The area of a parallelogram

1) Calculate the **area** of each **parallelogram**.



c) The **area of parallelogram** is 36 units<sup>2</sup>.

Calculate the size of the missing length and add it to the diagram.



$$36 = \text{base} \times 10$$

$$3.6 = \text{base}$$

The base should be marked parallel to the indicated line.

